Data Visualization with R

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Acknowledgement

Content Adapted directly from Fundamentals of Data Visualization by Claus O. Wilke, 2020

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Roadmap

- Basics & Visualizing Amounts
- Visualizing Distributions
- Visualizing Proportions
- Visualizing associations between variables
- Visualizing trends over time
- Visualizing uncertainty
- Strategies to avoid (if time permits)

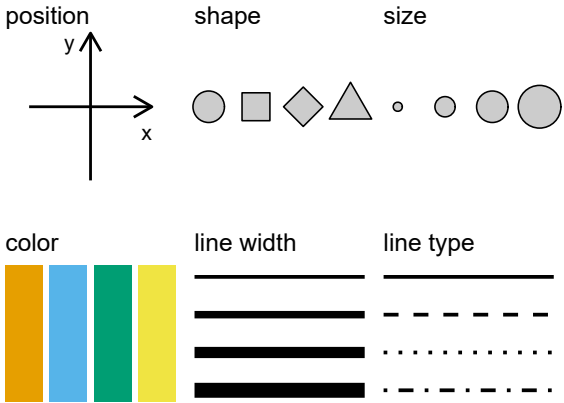
- For each topic:
  - Brief lecture
  - Hands-on in ggplot2

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Basics

Common Aesthetics

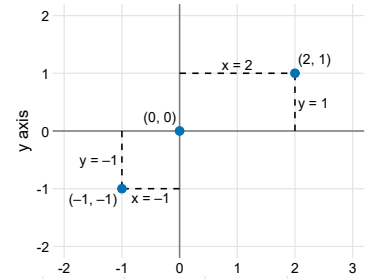
- Multiple aesthetic choices
  - Some convey only qualitative information (shapes)
  - Some convey quantitative information (position, size, color [qual and quant])

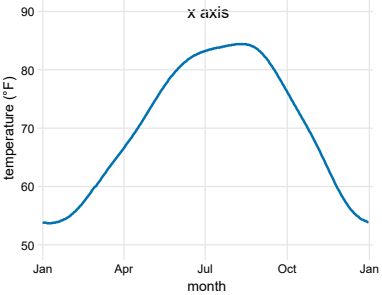


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Position

- Cartesian coordinates
  - 2D: x and y
  - Specify range
  - Units (label)
  - Scales should match if x and y are the same unit

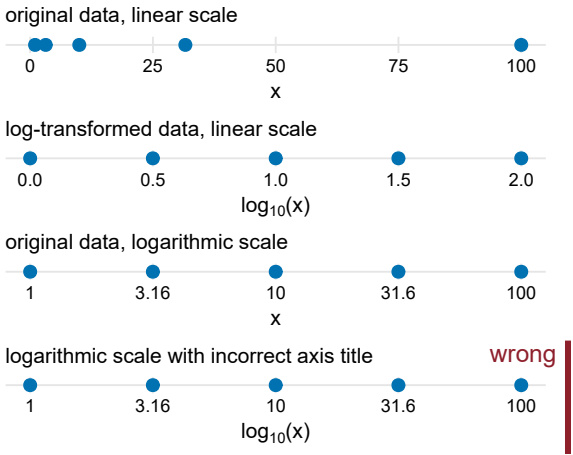




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Position

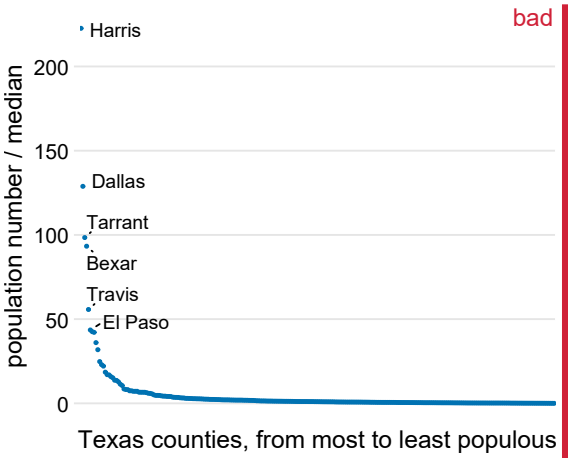
- Non-linear scales
  - Logarithmic
  - Square root
  - Other transformations



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Position

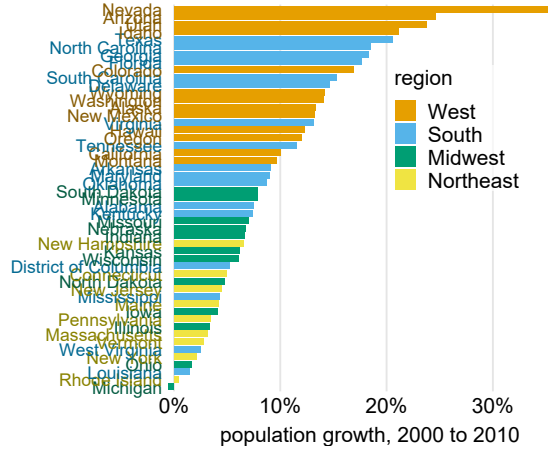
– Ratios (Odds, Relative Risk) best displayed on log scale



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Color

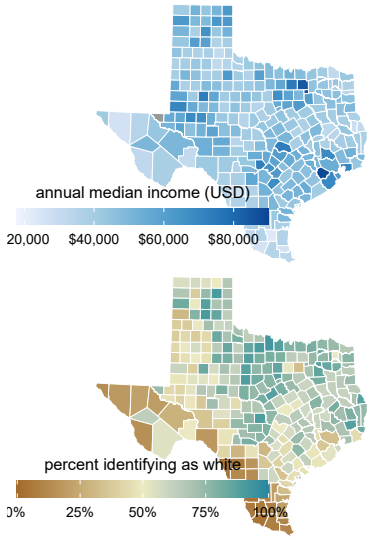
- Convey both qualitative and quantitative information
  - Qualitative: categorical variables
  - Quantitative: continuous variables
  - Qualitative scales should be distinct and easily distinguishable



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Color

- Quantitative color scales
  - Sequential: single hue, light to dark for intensity
  - Diverging: two hues, light to dark for difference from a midpoint



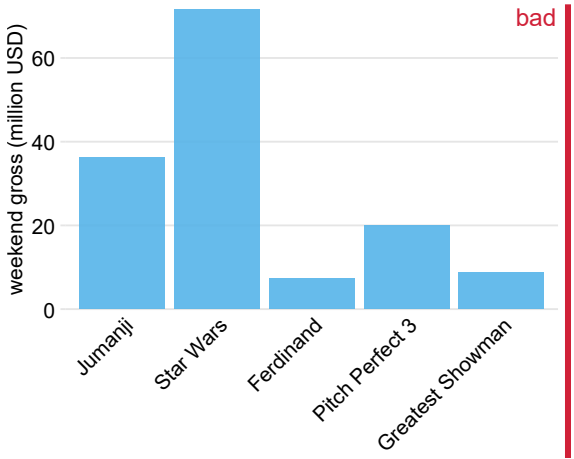
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Visualizing Amounts and Proportions

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Bar plots

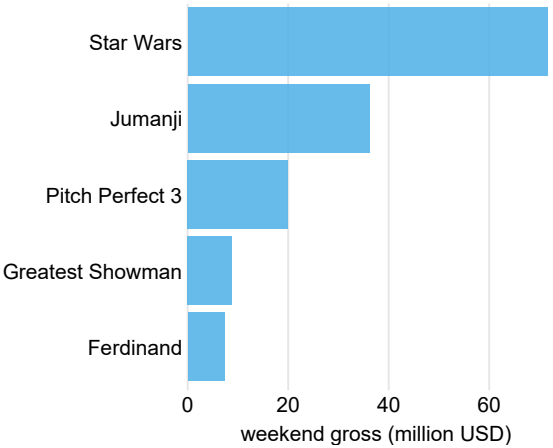
– Appropriate for continuous-by-categorical data



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Bar plots

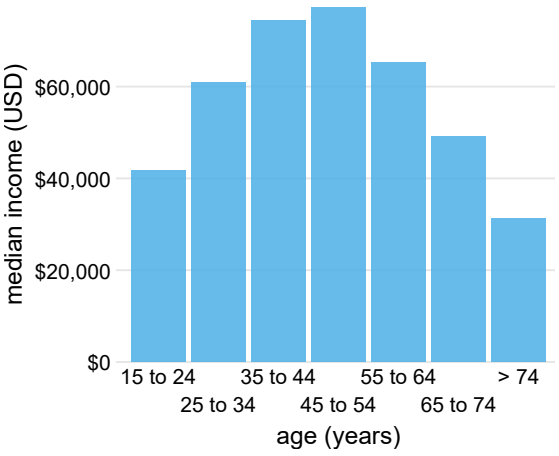
- Order matters (ascending or descending)
- Horizontal bars
  - Often allow for longer labels
  - Fit better on standard pages
  - Worse on standard slides



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Grouped bar plots

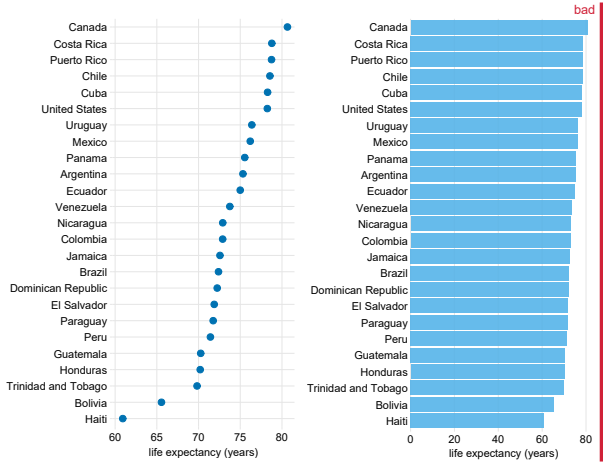
- Appropriate for continuous-by-categorical-by-categorical data
- Order matters (ascending or descending)
- Grouping matters - Bars you want to compare should be adjacent



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Dot plots

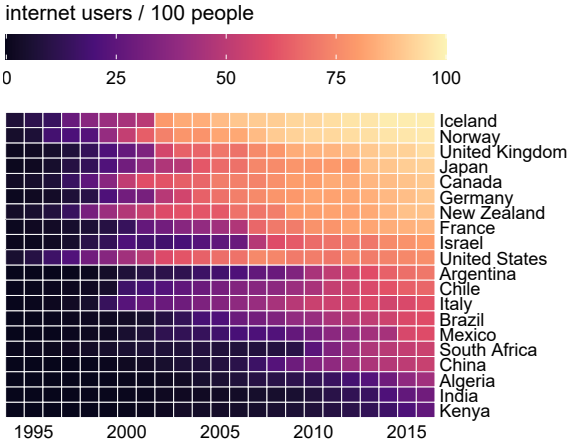
- Appropriate for continuous-by-categorical data
- Use when you do not want an axis to start at zero
- Order still matters



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Heatmaps

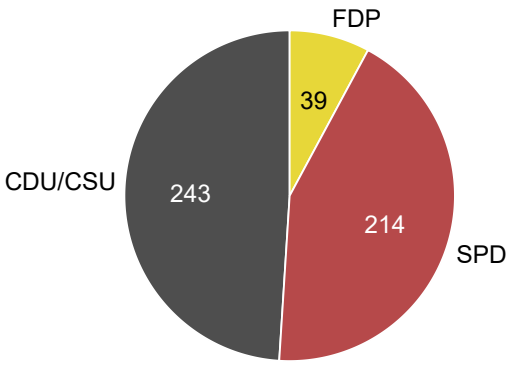
- Appropriate for continuous-by-categorical-by-categorical data
- Map continuous values to color not position
- Best when there are many groups in each category and you are implying broad trends not precise values
- Order still matters
- Use sequential or diverging color scales that are colorblind safe



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Pie charts

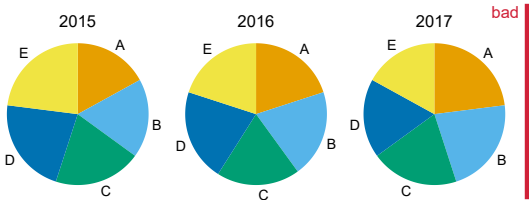
- Slice area = share of the whole
- Most useful with few categories and large percentages
- Works when the key message is a simple part-to-whole or majority comparison
- Example: SPD + FDP held a small majority over CDU/CSU

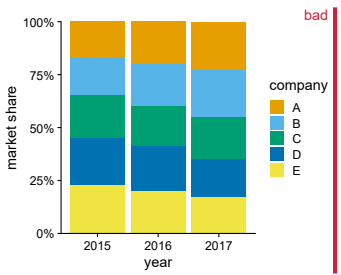


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When pie charts fail

- Five companies with similar shares (~20% each) over three years
- Pies: hard to compare shares within or across years
- Stacked bars: trends for A and E are visible, but middle bars shift across years
- Hard to compare anything but the bottom and top segments

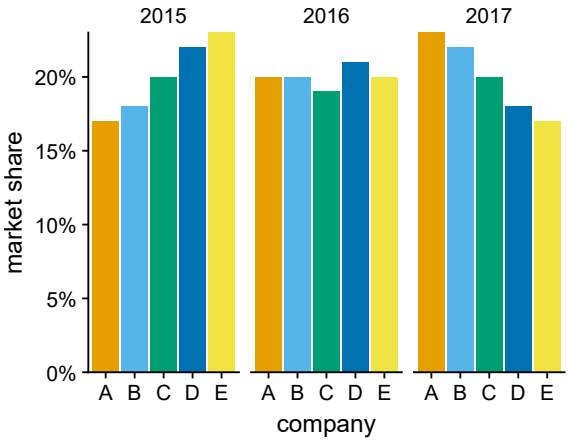




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Side-by-side bars

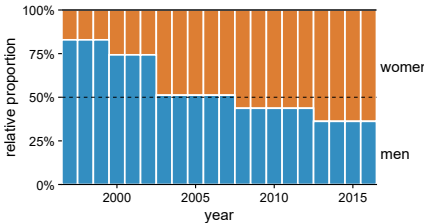
- Best choice for comparing parts across groups
- Shared baseline makes values easy to compare within a year
- A and B grew; D and E shrank
- Facet by year to see change over time

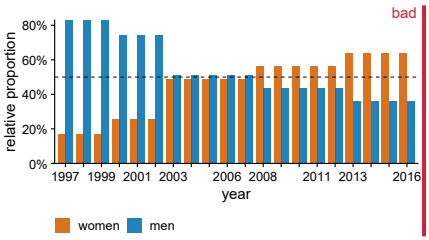


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Stacked bars with two categories

- Stacking works when there are only two parts per bar
- Women in the Rwandan parliament over time
- Dashed line at 50% shows when the majority flipped
- Side-by-side bars make the same data harder to read: the proportion is no longer visible as a whole





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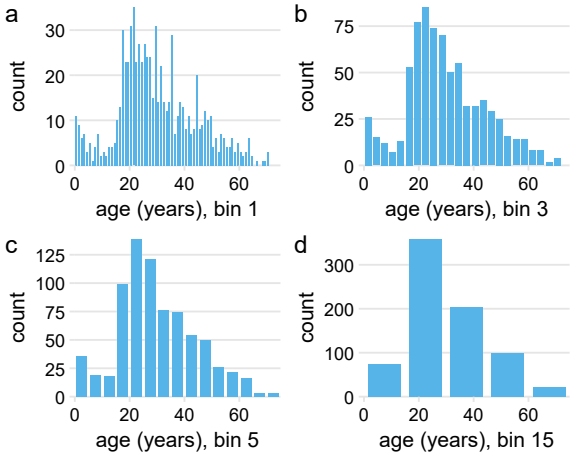
Visualizing Distributions

Distributions

- How a particular variable is distributed in a dataset
- Statistics: Mean, median, mode, standard deviation, skewness, kurtosis
- Visualizations: Histogram, density plot, boxplot, violin plot

Histograms

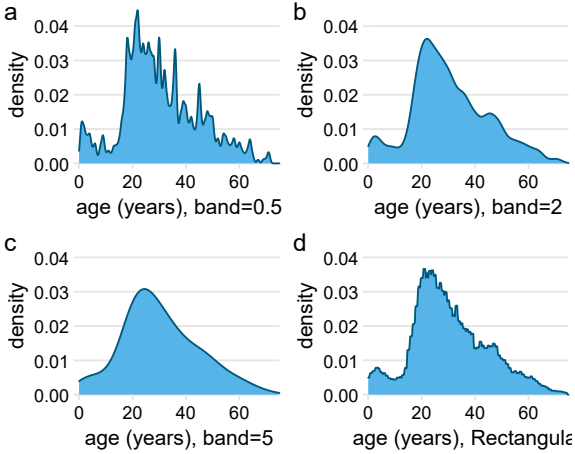
- Filled rectangles
- Bin = Width of rectangle = range of values in that bin
- Height of rectangle = number of observations or proportion of whole in that bin
- Bin width matters for storytelling



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Density plots

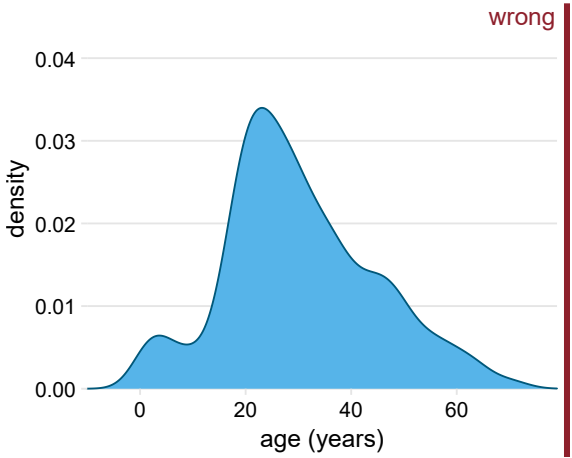
- Curve estimated from the data - smooth version of a histogram
- Area under the curve = 1
- Height of curve = density of observations at that value
- Density interpretation: Probability of observing a value in a given range = area under the curve for that range
- Bandwidth matters for storytelling



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Density plots

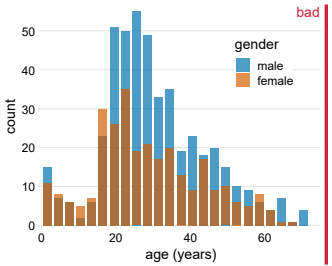
– Pitfall: smooth across gaps in data or where there are no observations.

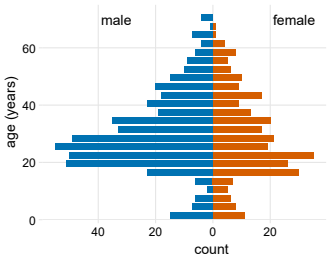


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Two distributions

- Avoid stacked histograms - hard to compare values
- Bars start at different points
- Use side-by-side histograms or density plots instead

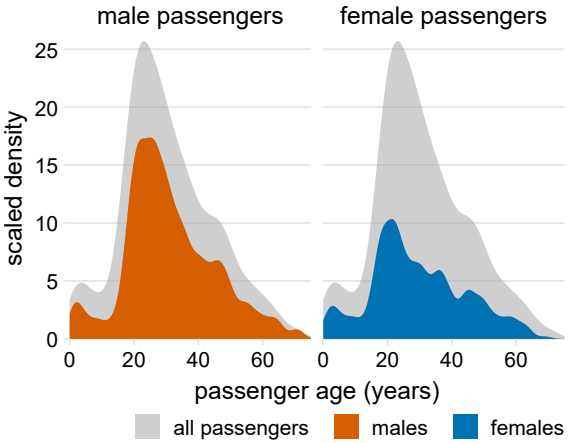




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Two distributions

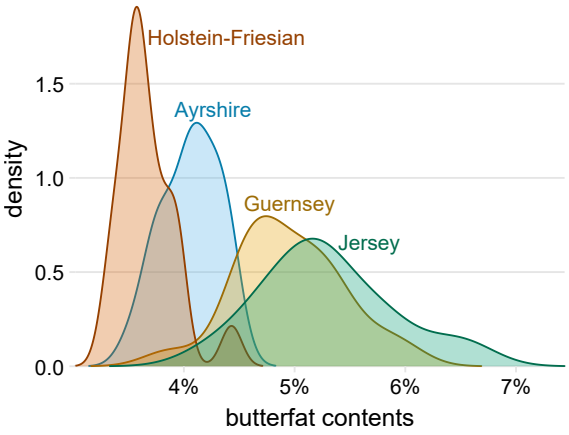
- Avoid stacked histograms - hard to compare values
- Bars start at different points
- Use side-by-side histograms or density plots instead



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More than two distributions

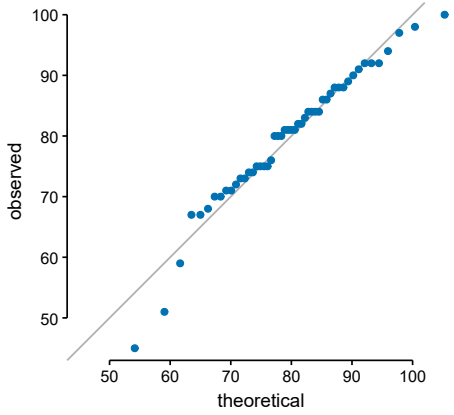
- Density plots extend nicely to more than one distribution
- Use color to separate distributions
- Transparency can help with overlapping distributions



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Quantile–quantile (Q-Q) plots

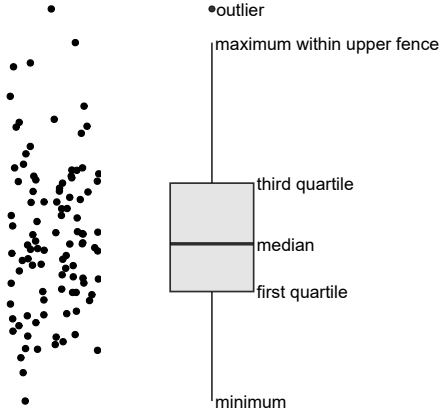
- Compare observed data to a reference distribution (usually normal)
- Rank each point, then plot its observed value vs. the value expected at that rank
- Diagonal line = x equals y
- Points on the line: data follow the reference distribution



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Boxplots

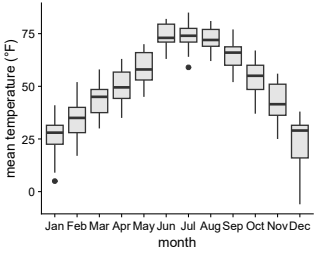
- Divide the data into quartiles
- Line = median; box = middle 50% of the data (interquartile range)
- Whiskers extend to the most extreme points within the fences (1.5 × box height)
- Points beyond the fences = outliers

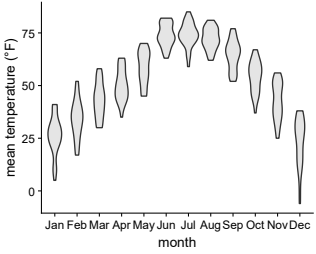


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Many distributions at once

- Boxplots: compact, easy to compare; show skew (e.g., December)
- Violin plots: mirrored density; reveal bimodality a boxplot hides (e.g., November)
- Violins inherit the shortcomings of density estimates

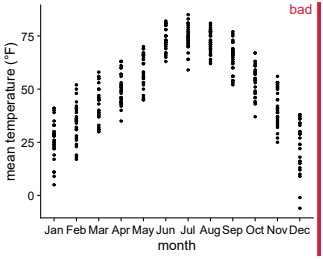


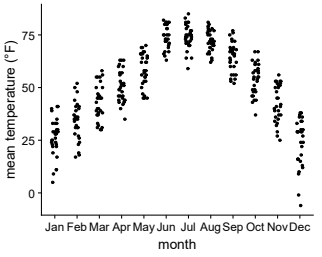


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Plotting all the points

- Strip chart: show every observation as a dot
- Overplotting hides which temperatures are most common
- Jittering spreads points along the *x* axis to show density
- Good option when the data are too sparse for violins



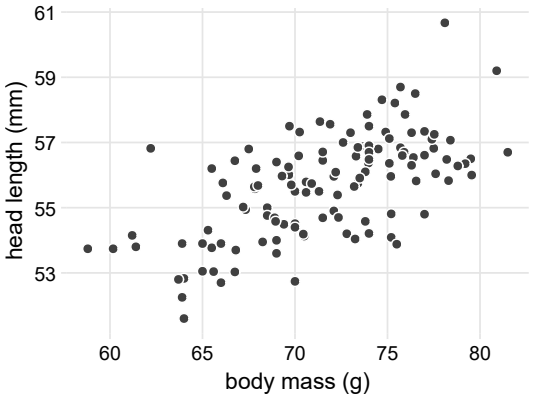


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Visualizing Associations

Scatter plots

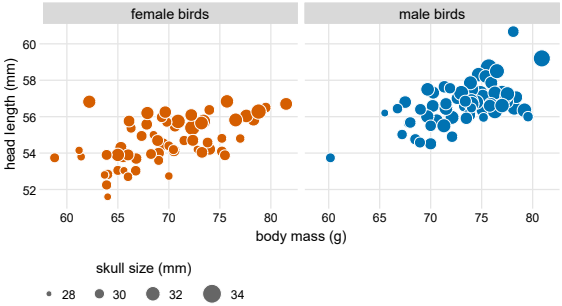
- Appropriate for two continuous variables
- Each point is one observation
- Look for direction, strength, and shape of the relationship
- Example: head length vs. body mass in blue jays



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Bubble charts

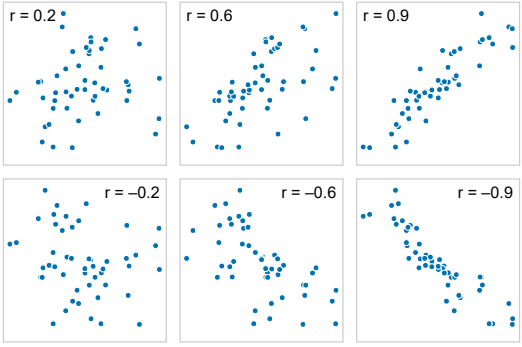
- Map a third continuous variable to point size (here, skull size)
- Color or facets can add a categorical variable (here, sex)
- Differences in bubble size are hard to see



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Correlations

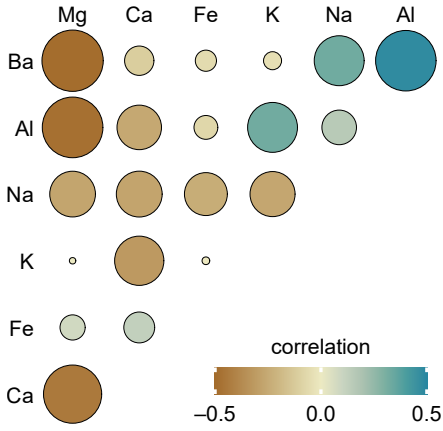
- Correlation coefficient r ranges from -1 to 1
- Sign = direction; magnitude = strength
- r = 0 means no linear association
- Same x and y values in every panel, only the pairing changes



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Correlograms

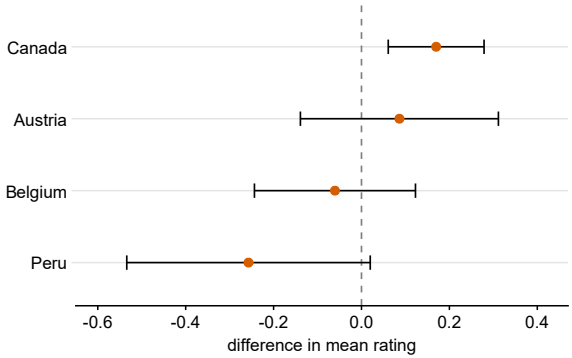
- Summarize many pairwise correlations at once
- Color = direction and strength; circle size = magnitude
- Size de-emphasizes correlations near zero
- Pitfall: abstract, and hides the underlying data



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Summary of differences across groups

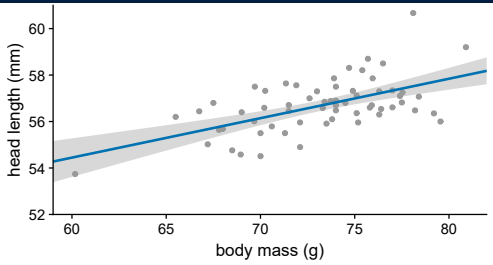
- Point = estimate; error bar = confidence interval
- Say what the bars represent (here, a 95% CI)
- Must select a reference group – here, chocolate from the US
- Dashed line at 0 = no difference from the reference group
- Could use aesthetics to highlight intervals that cross zero

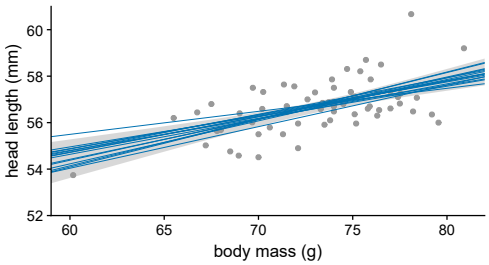


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Using fitted lines to see associations

- Confidence band: range of plausible fitted lines
- Confidence bands resemble alternative possible draws from the data
- Both show the same uncertainty in different ways
- Draws help avoid reading the band as exact

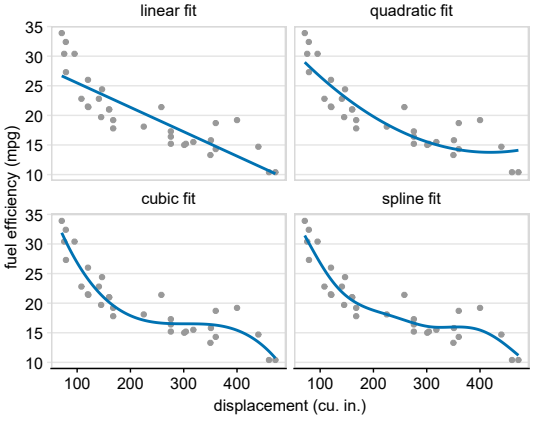




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Using fitted lines to see associations

- Same data, four different fits
- Linear: simple but may miss curvature
- Quadratic and cubic: bend, but can behave oddly at the ends
- Spline: flexible; follows the data most closely
- Choice of model can increase complexity and highlight storytelling



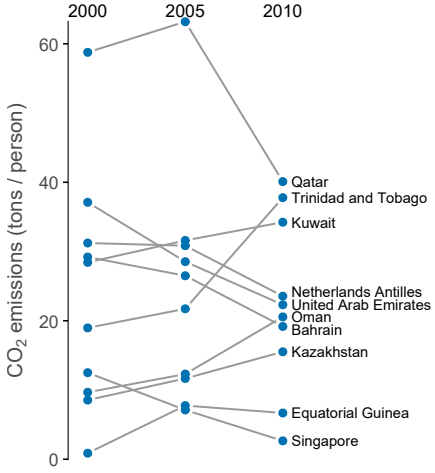
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Time Series and Trends

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Time Series

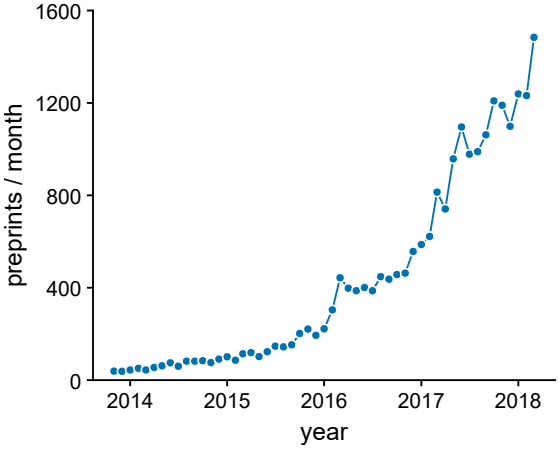
- Time on the *x* axis, a measurement on the *y* axis
- Connecting dots with lines shows order and trend
- The lines are a guide to the eye, not data
- Slopegraph: a few cases over a few time points
- Slope shows the size and direction of change
- Example: CO_2 emissions per person, 2000–2010



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Individual time series

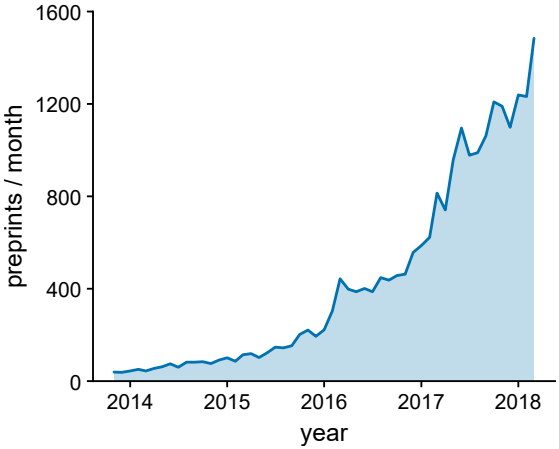
- Time on the *x* axis, a measurement on the *y* axis
- Connecting dots with lines shows order and trend
- The lines are a guide to the eye, not data
- Example: monthly bioRxiv preprint submissions



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Line and area plots

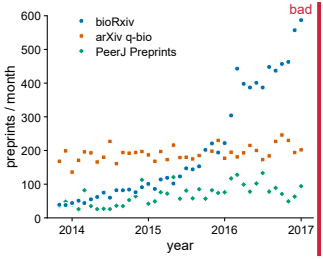
- Shading under the line emphasizes the magnitude, not just the trend
- Area plot appropriate when the y axis starts at zero
- Removing dots and adding area shading highlights overall trend relative to points

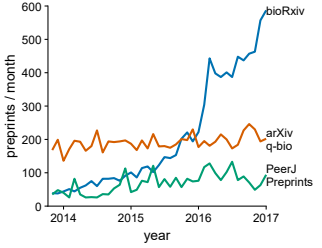


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Multiple time series

- Dots with a legend: hard to match colors and shapes to series
- Direct labels at line ends are easier to read
- Lines make each series' trend clear
- Order labels to match line ends

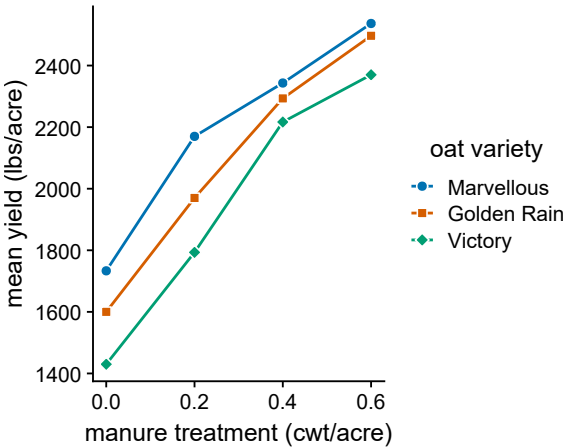




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Dose–response curves

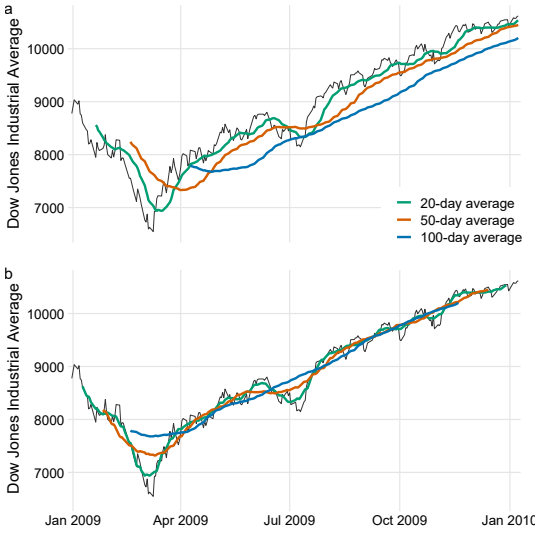
- The same line-plot approach works when x is not time
- x = treatment level (manure), y = mean response (oat yield)
- One line per group (variety) makes comparisons easy
- Use distinct colors and shapes for each group



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Moving averages

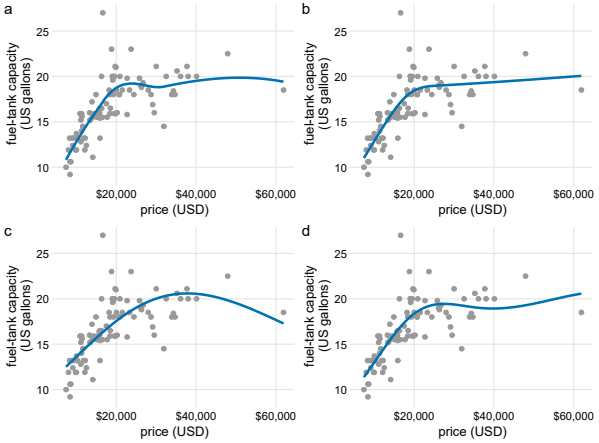
- Smooth noisy time series by averaging over a window
- Window length sets the scale of fluctuations that remain
- Trailing average (top) lags the data; centered average (bottom) does not
- Limits: shorter than the original series, and still wiggly



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Smoothers

- LOESS and GAMs fit smooth curves to scatter data
- Panel a is LOESS; panels b–d use different GAM settings
- Results depend heavily on the smoother and its parameters
- Try several choices before trusting a trend



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